Hyper Dimensional Probability - Python project

submit by:

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Part 1.Dataset Overview:

The dataset contains 303 samples and 14 columns in total. The target column (Label) is named target, and it indicates the presence of a disease. The distribution of the target variable in the dataset is relatively balanced, comprising 165 samples of sick patients (class 1) (54.45%) compared to 138 (45.55%) samples of healthy patients (class 0).
Features Description:
The independent features in the dataset are divided into 5 continuous columns and 8 categorical or binary columns. The continuous columns describe metrics such as age (age), resting blood pressure (trestbps), cholesterol level (chol), maximum heart rate (thalach), and the oldpeak index. Meanwhile, the categorical columns contain information about the patient's sex (sex), chest pain type (cp), fasting blood sugar levels (fbs), resting ECG results (restecg), exercise-induced angina (exang), slope (slope), number of major vessels (ca), and the thal index.

Data Preprocessing and Cleaning:
A comprehensive review of the dataset's clinical metrics revealed inconsistencies when compared to the original Cleveland dataset documentation. Specifically, values of ca = 4 (observed in 5 samples) and thal = 0 (observed in 2 samples) are incorrect data entries that originally mapped to missing values (NaNs). To ensure robust analysis and maintain data integrity, these 7 faulty entries were dropped from the dataset. Consequently, the effective sample size was reduced to 296 valid samples before any statistical transformation or modeling took place.

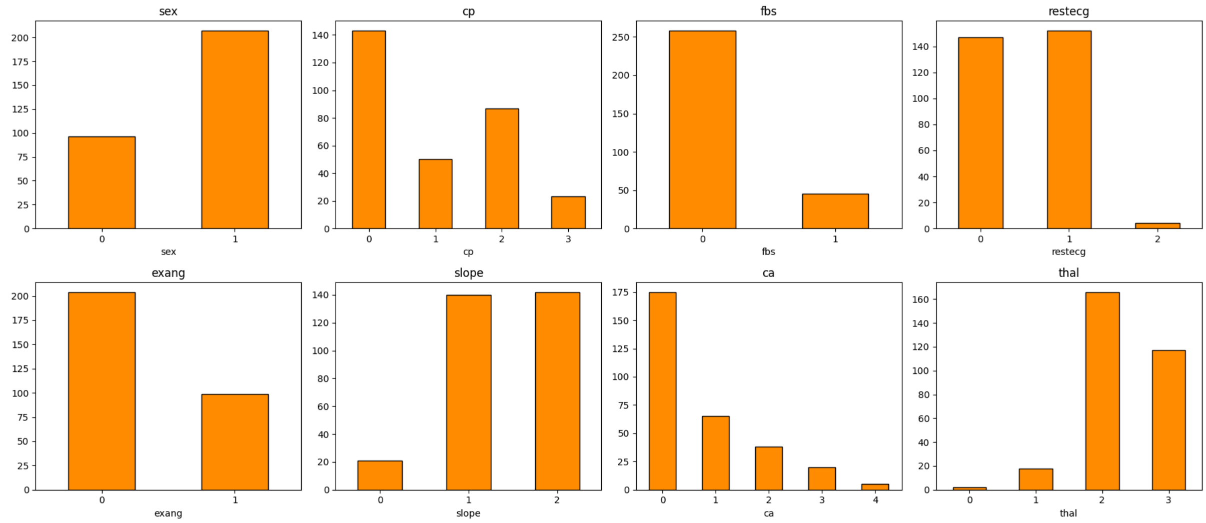
Prior to the normalization process, the continuous variables were characterized by significantly different scales. The data before the normalization process can be seen in the attached table:

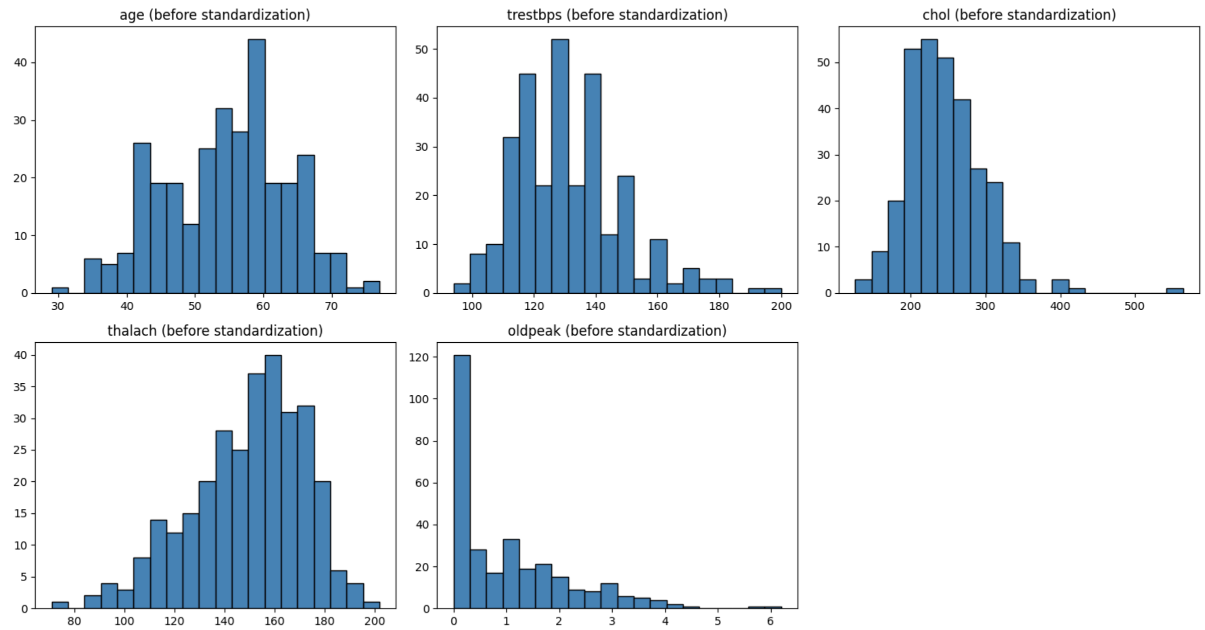
	mean	std
age	54.523649	9.059471
trestbps	131.604730	17.726620
chol	247.155405	51.977011
thalach	149.560811	22.970792
oldpeak	1.059122	1.166474

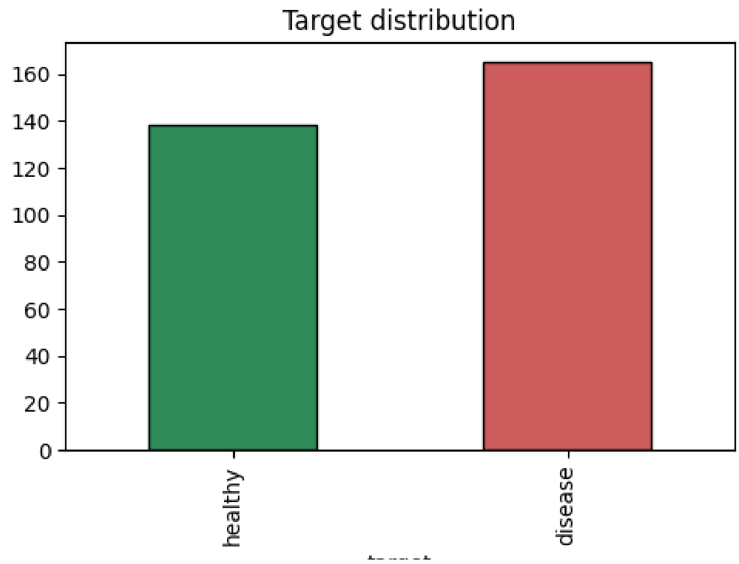
Therefore, we chose to normalize the data, and these are the values after normalization:

	mean	std
age	$3.840772 \cdot 10^{-16}$	1.001693
trestbps	$-6.001206 \cdot 10^{-16}$	1.001693
chol	$-2.340470 \cdot 10^{-16}$	1.001693
thalach	$1.680338 \cdot 10^{-16}$	1.001693
oldpeak	$-9.601929 \cdot 10^{-17}$	1.001693

Categorical Data Distribution: In addition, we examined the categorical variables to check the data distribution. For example, the **gender** distribution in the dataset is unbalanced, with 207 samples belonging to category 1 and 96 samples belonging to category 0. Furthermore, the vast majority of patients (258 samples) have a classification of 0 in the **fbs** column (compared to 45 in classification 1), and in 204 of the samples, no exercise-induced angina was observed (**exang** at classification 0). These distributions and others, along with the rest of the metrics, are presented here as images:


sex
cp
fbs
restecg
exang
slope
ca
thal


age (before standardization)
trestbps (before standardization)
chol (before standardization)
thalach (before standardization)
oldpeak (before standardization)


Target distribution
healthy
disease

Several features within the dataset (e.g. cp, thal, and restecg) are categorical in nature, despite being initially encoded as integers. Since these numerical representations lack ordinal significance, treating them as continuous variables would introduce artificial spatial relationships and distort distance metrics. To guarantee that subsequent geometric analyses (such as PCA and intrinsic dimensionality estimation) rely on valid distance metrics, one-hot encoding was applied to all categorical features. This transformation effectively expanded the spatial dimensionality of the data from 13 to 20 dimensions.
At the end we have 4 datasets:

File name	Dataset content
heart.csv	the original dataset
heart_clean.csv	the data after we deleted the lines that had missing or non-allowed cells
heart_standartized	took the clean data and normalizing the numerical features
heart_features.csv	the final data after we splited the categorial features

Part 2. Principal Component Analysis (PCA):

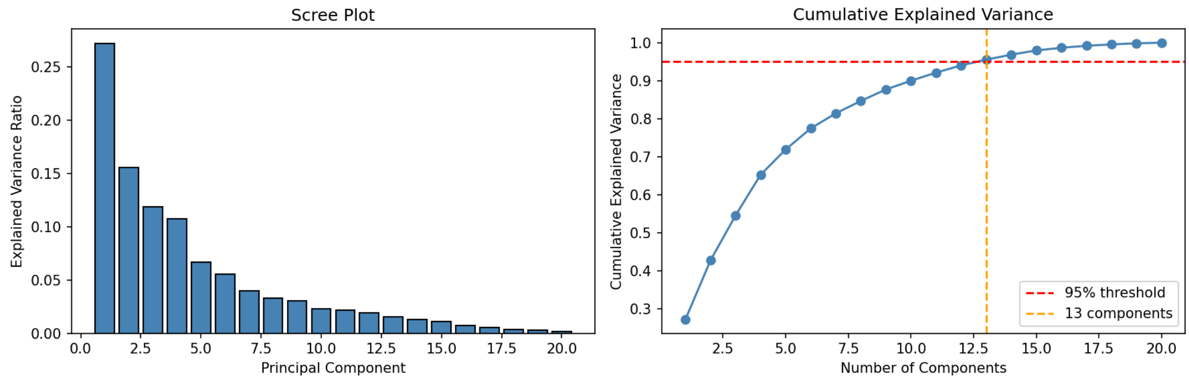
2.1 motivation:

PCA finds the orthogonal directions of maximum variance in the data. It allows us to answer two key questions:

1. How many dimensions are truly needed? i.e., what is the effective linear dimensionality?
2. Which features drive the variance? i.e., what are the dominant directions in feature space.

2.2 Effective Dimensionality:

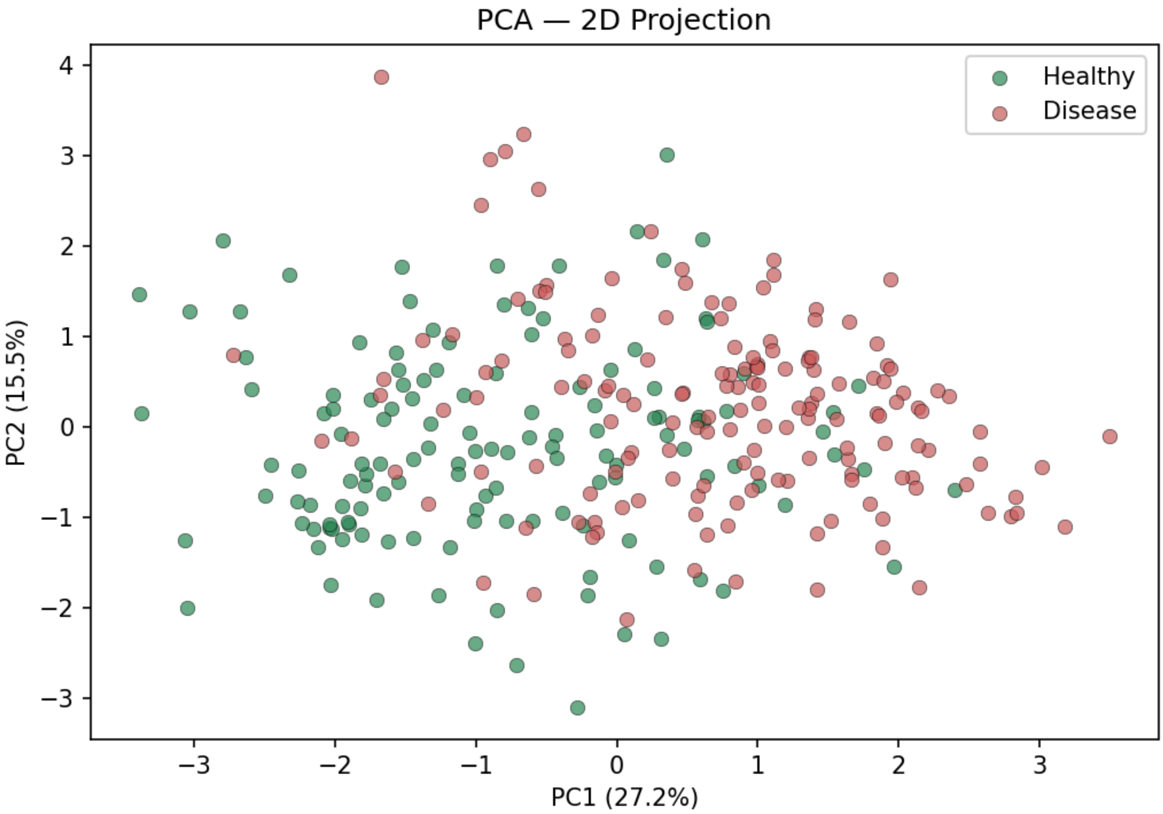
PCA was applied to the full 20-dimensional feature matrix. The figure below shows the explained variance per component (scree plot) and the cumulative explained variance.

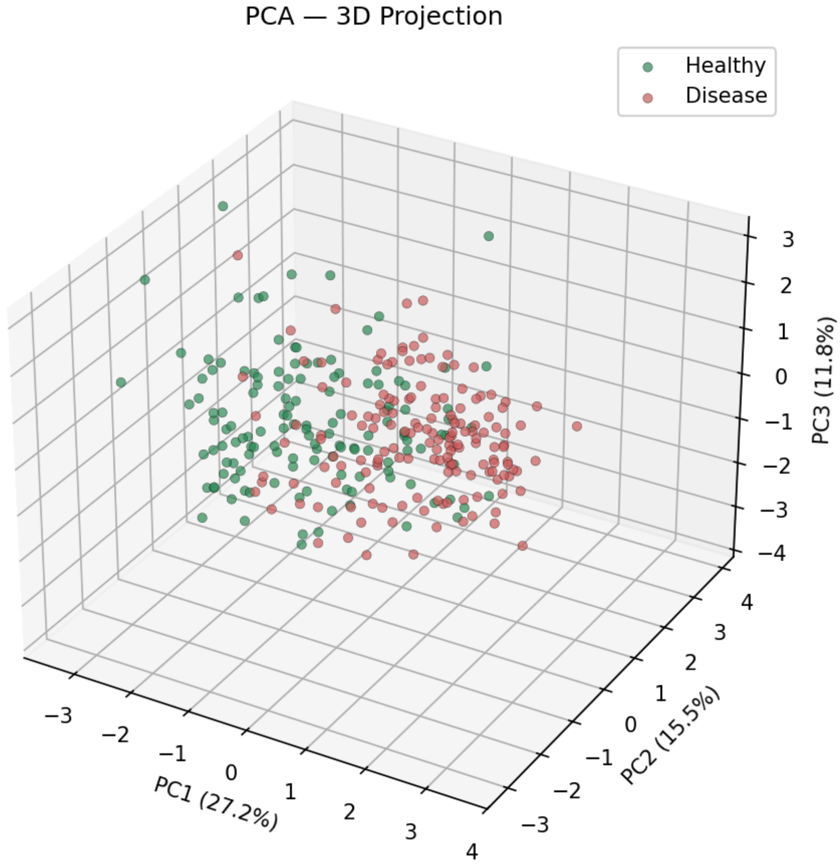


In the left graph we can see the variance ratio of each component.
In the right graph we can see the cumulative function of the variance
Finding: A total of 13 principal components are required to explain 95% of the cumulative variance. This suggests that the dataset does not compress efficiently in a linear fashion, as the variance is widely distributed across multiple dimensions rather than concentrated in a few. Specifically, the first five components account for 27.2%, 15.5%, 11.8%, 10.7%, and 6.6% of the variance, respectively. This lack of a single dominant axis is characteristic of heterogeneous datasets, where continuous clinical measurements are intertwined with multiple binary indicators.

2.3 2D and 3D Projections

The data was projected onto the first two and three principal components to visualise the class separation.

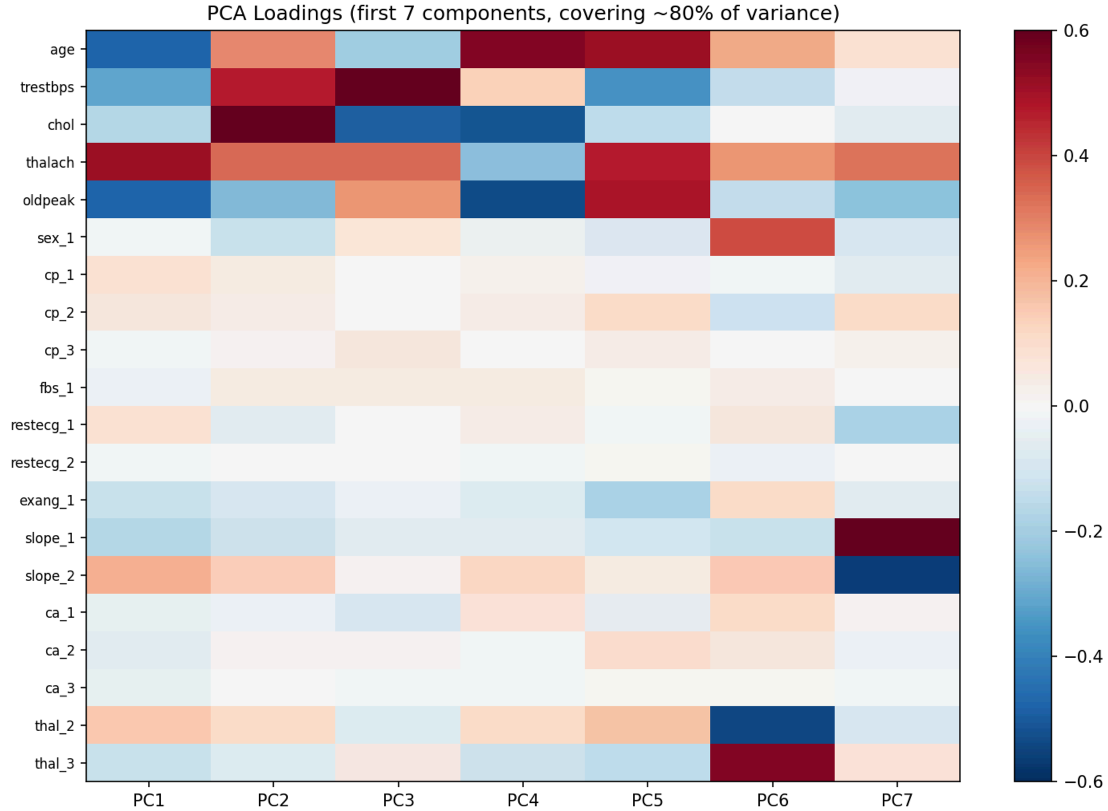




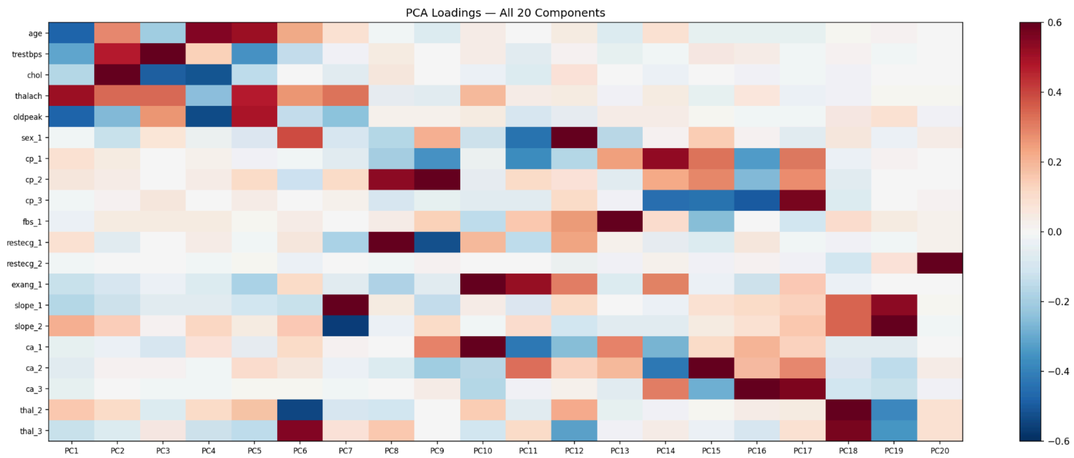
Finding: The two classes (healthy vs disease) show partial but imperfect separation in the first two principal components. This confirms that some linear structure exists in the data, though it is not sufficient for clean linear separation — consistent with the distributed variance observed in the scree plot.

3.4 Feature Loadings

The heatmap below shows the contribution (loading) of each original feature to the first 7 principal components. Seven components were chosen because they collectively explain approximately 80% of the total variance — a standard threshold that balances coverage with visual readability.



we also created one for all the components but it must harder to see and because of this we created



Finding: PC1 is dominated by `thalach` (loading: +0.509), `oldpeak` (−0.481), and `age` (−0.474). This axis captures an overall cardiac fitness gradient: patients with high maximum heart rate score positively, while patients with high ST depression and older age score negatively — a clinically coherent direction that aligns closely with disease severity.

PC2 is driven by `chol` (+0.650) and `trestbps` (+0.473), representing a metabolic risk axis independent of the fitness gradient captured by PC1. The two axes together explain 42.7% of the total variance.

The loadings heatmap confirms that no single feature dominates all components — the variance is genuinely distributed across the feature space, which is consistent with the high effective dimensionality (13 components for 95%) observed in the scree plot.

Part 3.Intrinsic Dimensionality Estimation

3.1 Motivation:

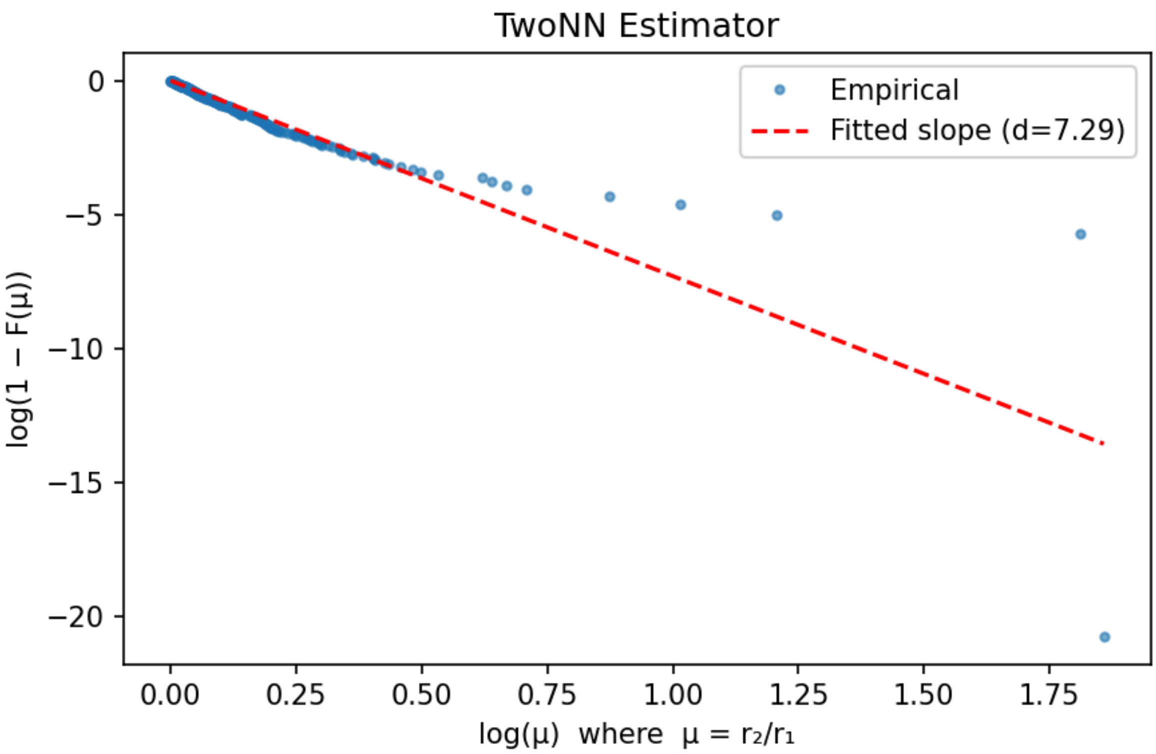
PCA revealed that 13 linear components are needed to explain 95% of the variance. However, PCA only captures linear structure. A more fundamental question is: **what is the intrinsic dimensionality of the manifold on which the data lies?**

Consider a cylinder embedded in 3D space — PCA would assign it 3 dimensions, but its surface is inherently 2-dimensional. Intrinsic dimensionality estimators attempt to recover this true geometric dimension from the local neighbourhood structure of the data.

Three complementary estimators were applied, each with different assumptions and sensitivity to noise.

3.2 TwoNN Estimator

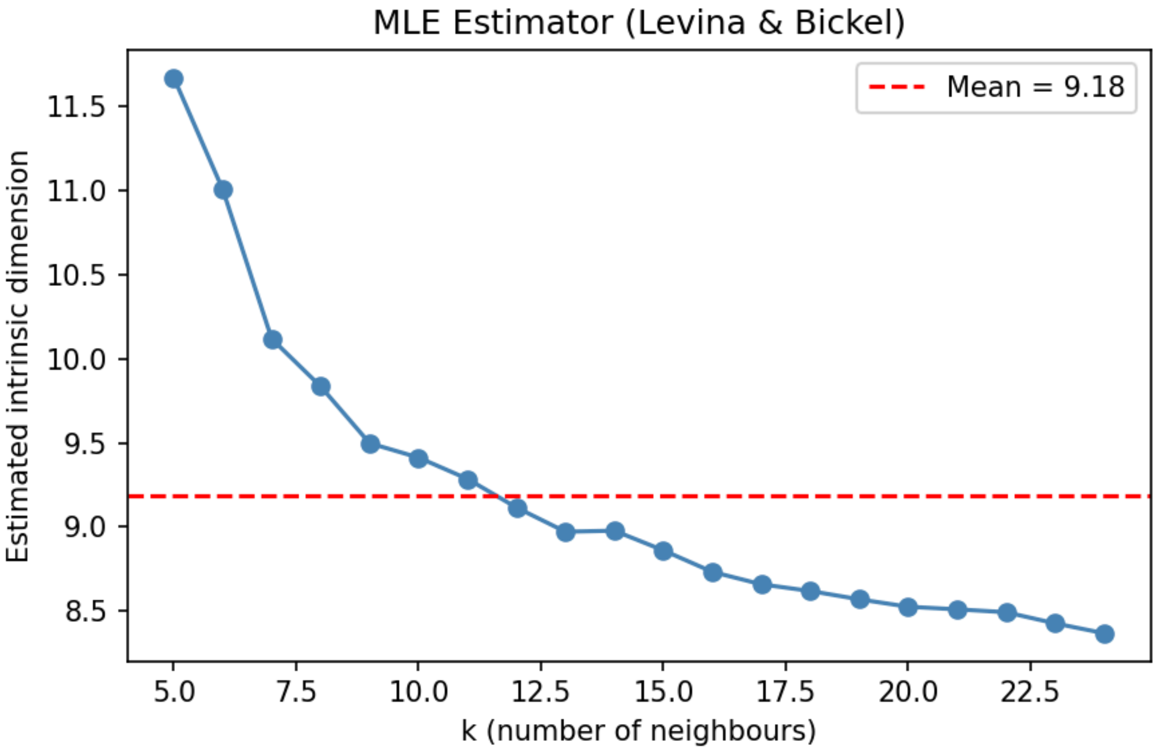
The TwoNN method (Facco et al., 2017) is based on a simple observation: for each point, if the ratio $\mu = r_2/r_1$ (distance to 2nd nearest neighbour divided by distance to 1st) follows a Pareto distribution, the shape parameter of that distribution equals 1/d. The intrinsic dimension is estimated as d = 1 / mean(log μ).



Finding (TwoNN): The plot shows log(1 − F(μ)) against log(μ), where each blue dot represents one patient. The x-axis measures how large the gap is between the 1st and 2nd nearest neighbours (larger log(μ) = bigger gap), while the y-axis shows what fraction of patients have an even larger gap. The red line is the theoretical Pareto fit whose slope equals the estimated dimension. The blue dots follow the red line closely, confirming that the Pareto assumption holds for this dataset. The estimated intrinsic dimension is **7.29**.

4.3 MLE Estimator (Levina & Bickel)

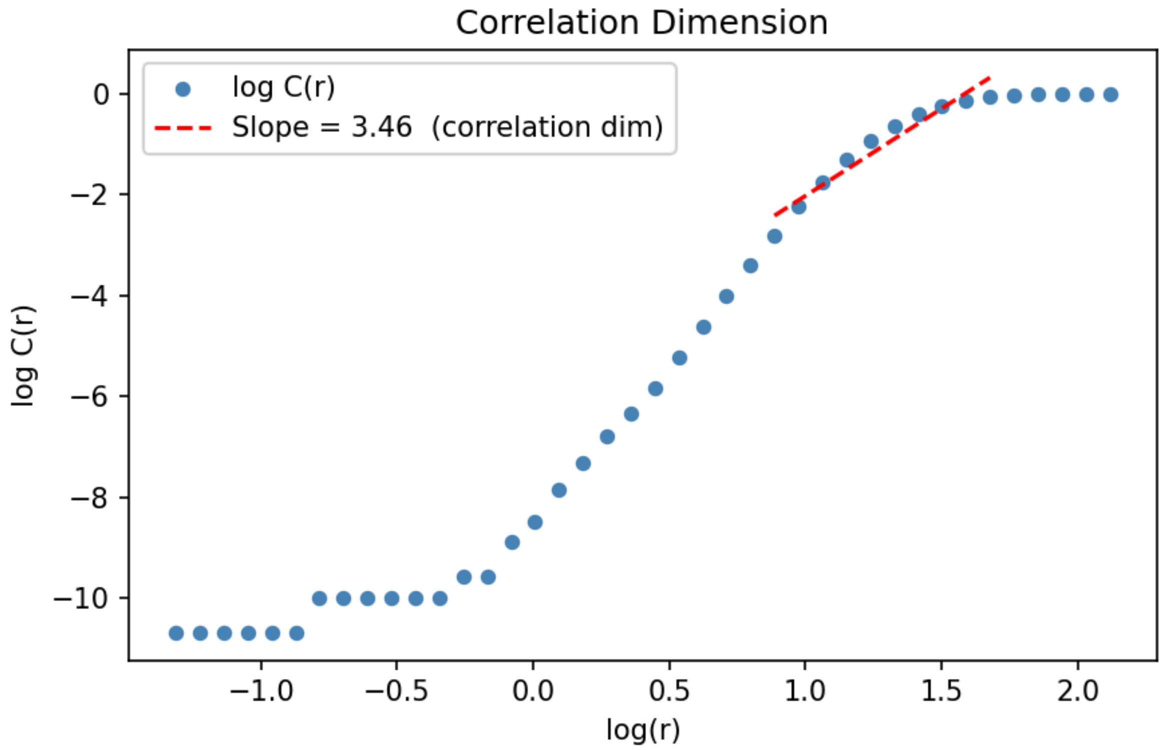
For each point, this estimator uses its k nearest neighbours and computes the maximum-likelihood estimate of the local dimension based on how distances grow with k. The estimate is computed across k = 5…24 and averaged for stability.



Finding (MLE): Each point on the plot represents the average estimated local dimension for a given k. The estimate is stable across k = 5…24, hovering around 9, which indicates the result is not sensitive to the choice of neighbourhood size. The mean estimate is **9.18**.

4.4 Correlation Dimension

This method counts the fraction of point-pairs within distance r for a range of radii. In a d-dimensional space, this count grows as r^d — so the slope of a log C(r) vs log(r) plot directly estimates d. The slope is fitted in the linear scaling region (5%–95% of pairs) to avoid boundary effects.



Finding (Correlation Dimension): The blue dots are the empirical log C(r) values and the red dashed line is the linear fit in the scaling region. The slope of the fit is **3.46**. This is lower than the TwoNN and MLE estimates because the correlation dimension measures global scaling behaviour of the entire point cloud, rather than local neighbourhood ratios. In our dataset, the presence of binary one-hot encoded features creates clusters of equal pairwise distances, which compresses the effective scaling range and leads to a lower estimate.

4.5 Summary

Method	Estimated Dimension
Ambient (total features)	20
PCA linear (95% variance)	13
TwoNN	7.29
MLE (Levina & Bickel)	9.18
Correlation Dimension	3.46

All three nonlinear estimators agree that the true intrinsic dimensionality is **substantially lower** than both the ambient dimension (20) and the PCA linear dimension (13). The convergence of TwoNN and MLE around 7–9 is the most reliable signal, as both methods measure local neighbourhood structure independently. The lower correlation dimension (3.46) reflects global distance clustering caused by binary features, and should be interpreted as a lower bound. Together, these results indicate that while the dataset does not suffer severely from the curse of dimensionality, it is not trivially low-dimensional either.

Part 4. Random Projections — Johnson-Lindenstrauss Lemma

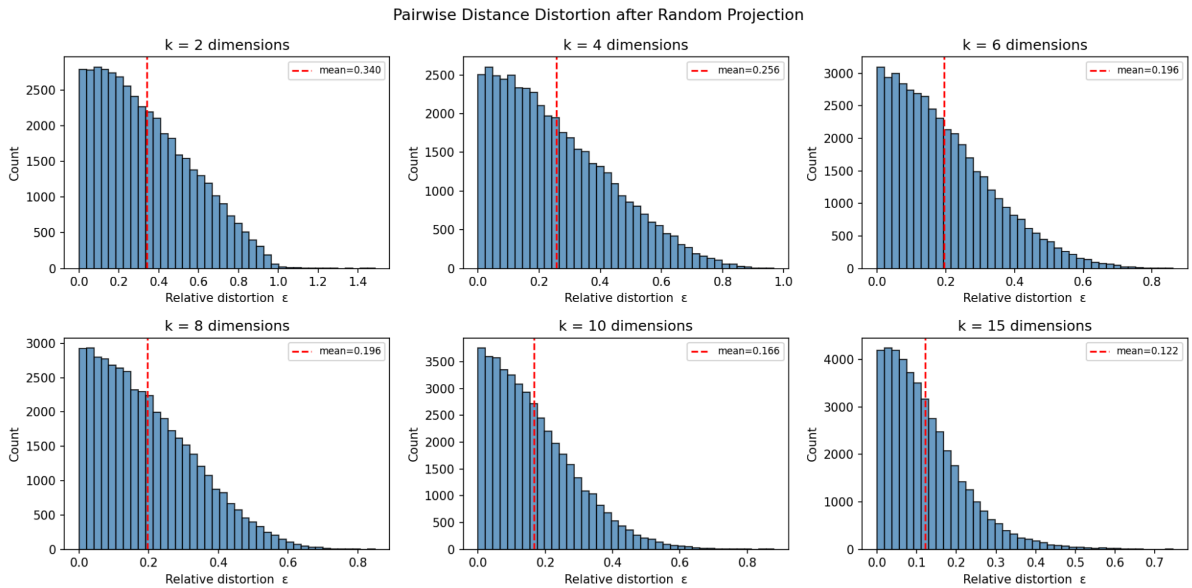
4.1 Motivation

The Johnson-Lindenstrauss (JL) lemma states that a set of n points in high-dimensional space can be projected to a much lower-dimensional space using a random linear map, while approximately preserving all pairwise distances. Formally, for any $\varepsilon > 0$, a projection to $k = O(\log n / \varepsilon^2)$ dimensions suffices to ensure that all distances are distorted by at most a factor of $(1 \pm \varepsilon)$.

This analysis tests the lemma empirically on our data: we project the 20-dimensional feature matrix to k = 2, 4, 6, 8, 10, 15 dimensions using random Gaussian matrices, and measure how well pairwise distances are preserved.

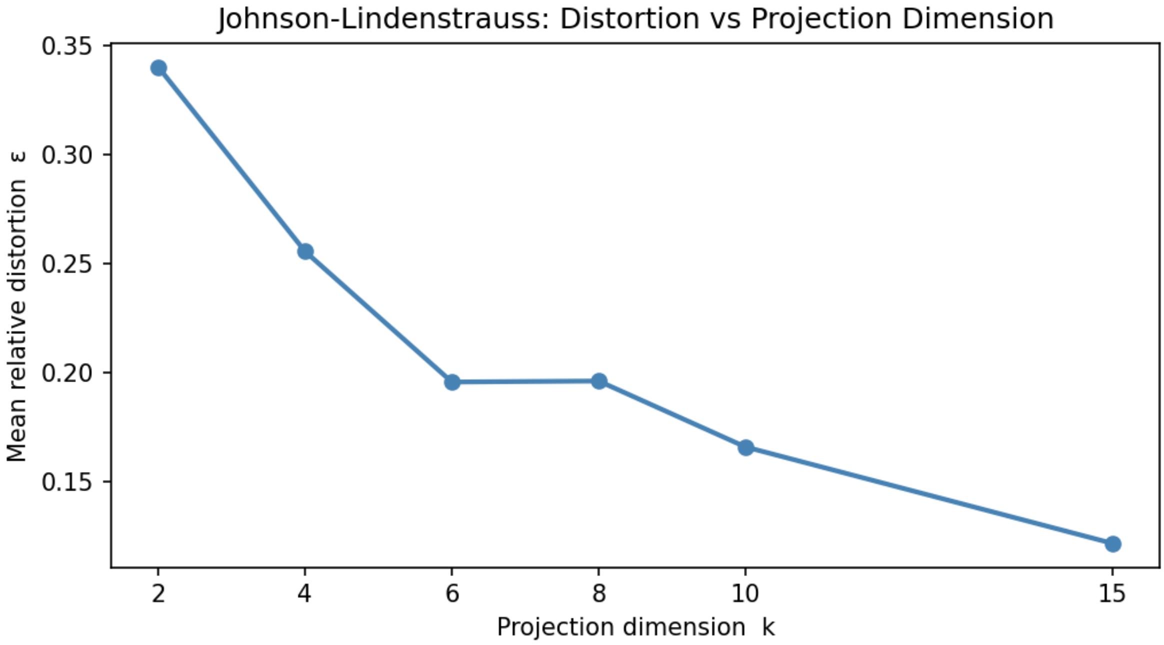
4.2 Distortion Distributions:

For each projection dimension k, the relative distortion is computed for all ~43,000 (296 chose 2) patient pairs as $\varepsilon = |d\_projected - d\_original|$ / d_original.



Finding: Each histogram shows the distribution of relative distortion across all ~43,000 patient pairs for a given projection dimension k. At k=2, the mean distortion is 34% — most distances are heavily distorted. As k increases, the distribution shifts left and narrows: at k=10 the mean distortion drops to 16.6%, and at k=15 to 12.2%. This confirms the JL lemma empirically — more dimensions yield better distance preservation.

4.3 Mean Distortion vs Projection Dimension



Finding: The mean distortion decreases monotonically as k increases, following the expected JL behaviour. Notably, at k=15 (75% of the original 20 dimensions) the mean distortion is already below 13%, demonstrating that most of the geometric structure of the dataset is captured in a much lower-dimensional random subspace. This is consistent with the intrinsic dimensionality estimates from Section 4, which suggested the data effectively lives on a ~7–9 dimensional manifold.

Part 5. Random Matrix Theory — Marchenko-Pastur Law

5.1 Motivation

When analysing the covariance matrix of a dataset, a fundamental question is: which eigenvalues represent true signal (genuine structure in the data), and which are simply noise artefacts that would appear even in a fully random dataset?

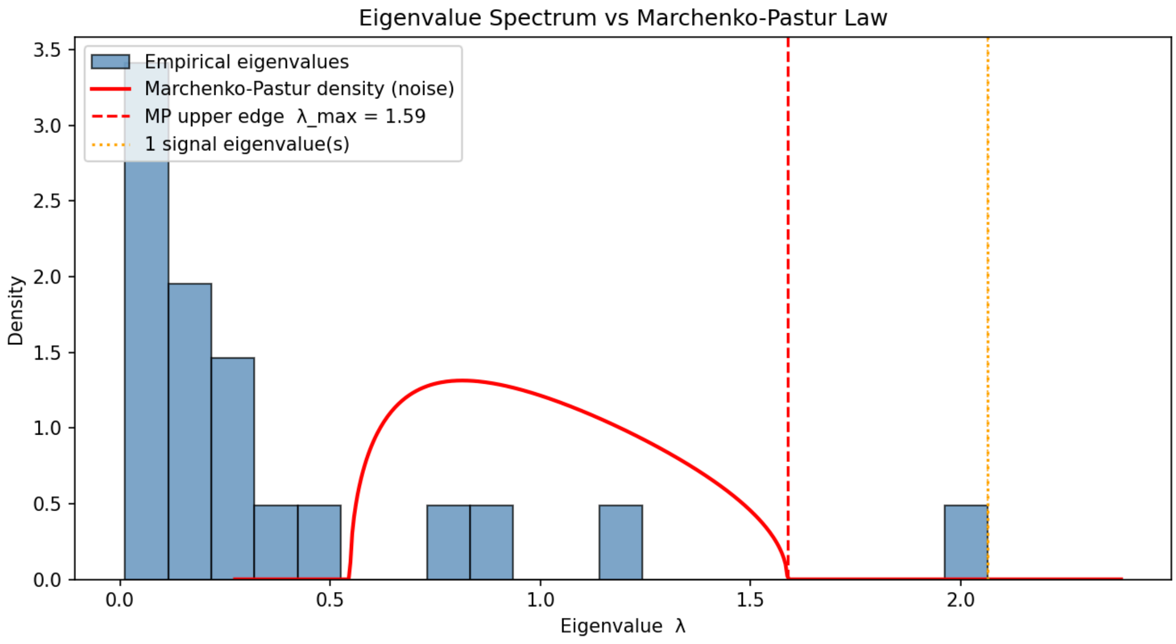
The Marchenko-Pastur law provides a theoretical answer. For a matrix of N samples and D features where entries are i.i.d. random variables, the eigenvalue distribution of the sample covariance matrix converges to a known density with support [λ_{min}, λ_{max}], where:

$\lambda_{min/max} = \sigma^2(1 \mp \sqrt{\gamma})^2$, with $\gamma = D/N$ (the aspect ratio)

Eigenvalues above **λ_{max}** cannot be explained by randomness alone — they carry genuine signal.

5.2 Eigenvalue Spectrum

For our dataset: N=296, D=20, $\gamma = 20/296 \approx 0.068$. The MP bulk spans [0.548, 1.587].



Finding: The histogram shows the empirical eigenvalue distribution of the sample covariance matrix. The red curve is the theoretical Marchenko-Pastur density for pure noise (γ=0.068, σ^2=1), and the red dashed vertical line marks its upper edge at λ_{max}=1.587. The orange dotted lines mark individual eigenvalues above this threshold.

Only **one eigenvalue** (λ=2.064) lies above the MP upper edge. This means that under a random matrix theory lens, only a single direction in the covariance space carries a statistically significant signal — corresponding to PC1 identified in Section 3. All remaining 19 eigenvalues fall within the MP bulk and cannot be distinguished from noise.

This is a strong and somewhat surprising finding: despite the dataset having 20 features and an apparent linear dimensionality of 13 (PCA, 95% threshold), only one covariance direction is certifiably above the noise floor. This suggests that the majority of the variance captured by PCA components 2–13 may reflect structured noise or correlations among features rather than independent signal dimensions.

Part 6. Anomaly Detection

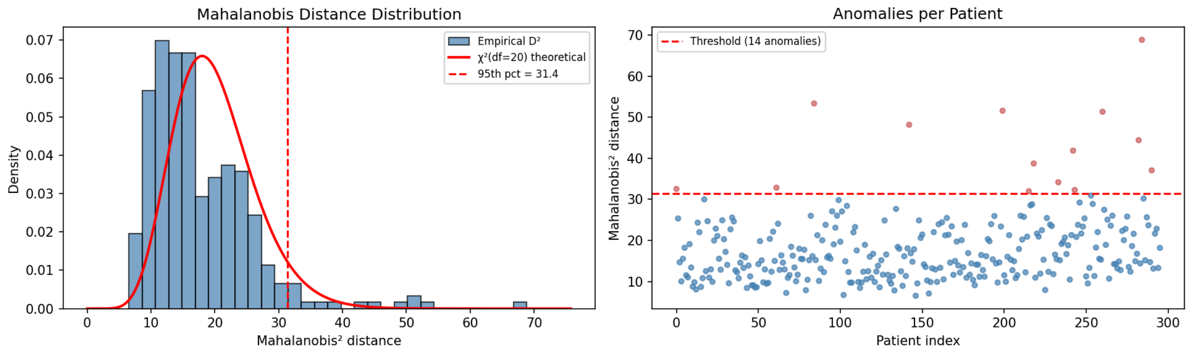
6.1 Motivation

In high-dimensional spaces, distance-based outlier detection becomes challenging due to concentration of measure — pairwise distances tend to equalize, making it harder to distinguish outliers from normal points. Two complementary methods are applied to identify patients that deviate significantly from the population distribution.

6.2 Mahalanobis Distance

The Mahalanobis distance measures how far each patient is from the population centroid, accounting for feature correlations and scale. Unlike Euclidean distance, it normalises for the shape of the data cloud. Under the assumption of multivariate Gaussian data, the squared Mahalanobis distance follows a $\chi^2(D)$ distribution, providing a principled threshold for anomaly detection.

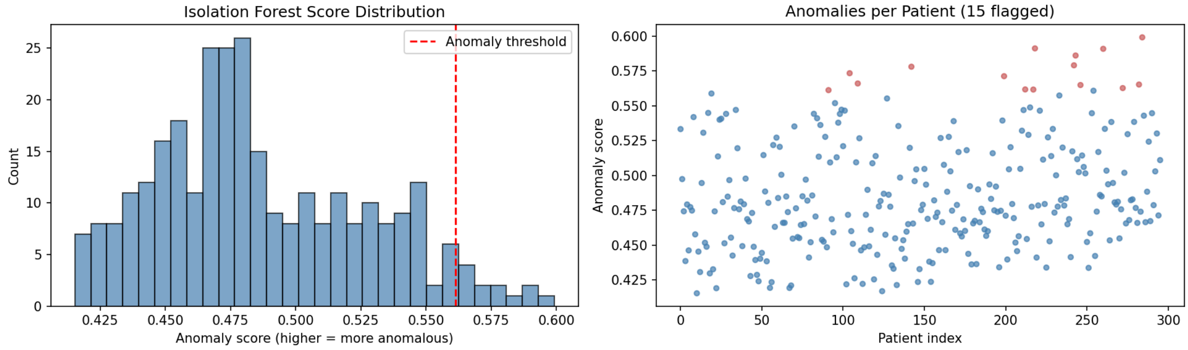
A Ledoit-Wolf shrinkage estimator was used for the covariance matrix to improve numerical stability with D=20 features.



Finding (Mahalanobis): The left plot shows the distribution of squared Mahalanobis distances against the theoretical $\chi^2(20)$ density. The empirical distribution has a heavier right tail than the theoretical curve, suggesting the data is not perfectly Gaussian — consistent with the presence of binary features. The right plot shows each patient's distance; 14 patients (4.7%) exceed the 95th percentile threshold of 31.4 and are flagged as anomalies (red points).

6.3 Isolation Forest

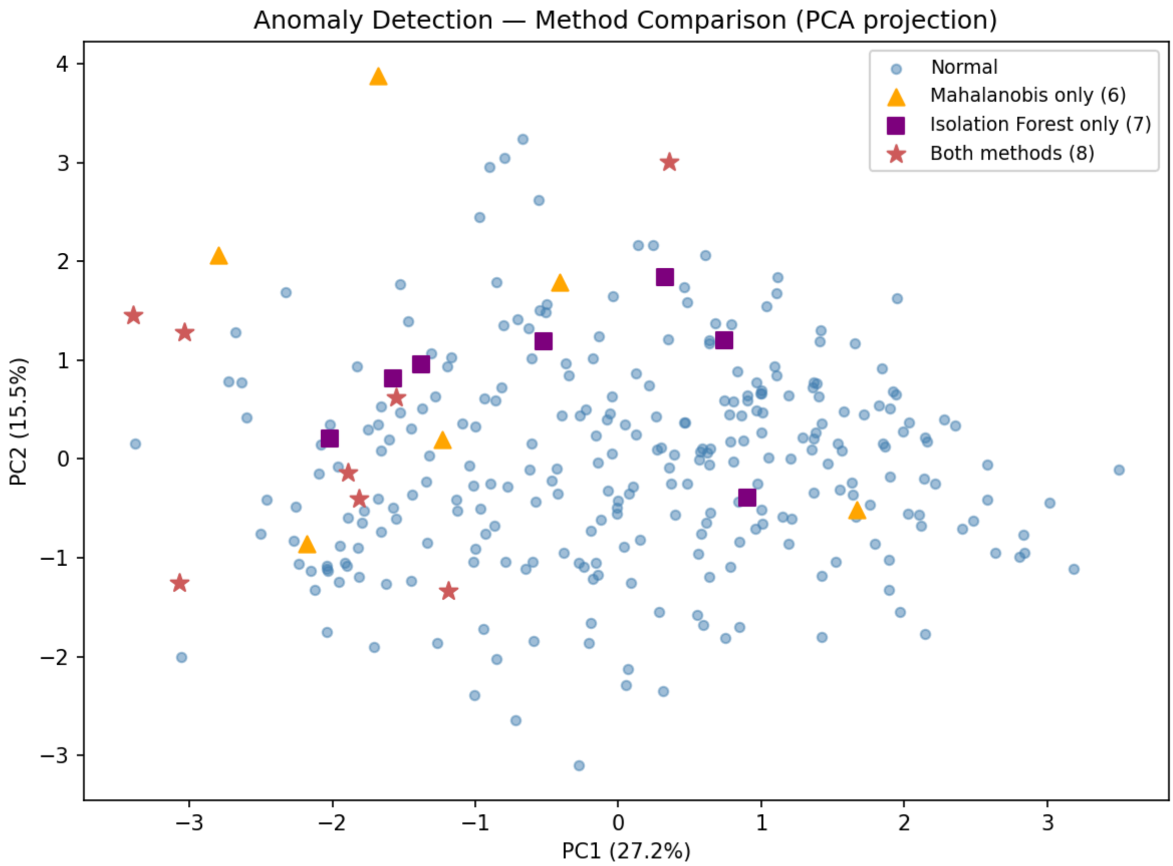
Isolation Forest detects anomalies by recursively partitioning the feature space with random splits. Points that are isolated in fewer splits receive a higher anomaly score. A contamination rate of 5% was set, consistent with the Mahalanobis threshold.



Finding (Isolation Forest): The left plot shows the distribution of anomaly scores. The threshold separating the bottom 95% from the top 5% is clearly visible. 15 patients (5.1%) are flagged as anomalies. The score distribution is roughly unimodal with a light tail, indicating that extreme outliers are rare.

6.4 Comparison

The figure below projects both anomaly results onto the first two PCA components to compare which patients each method flags.



Finding (Comparison): Of 22 unique anomalies flagged across both methods, 8 patients are identified by both — these represent the most robust outliers. 6 patients are flagged only by Mahalanobis (globally distant from the centroid but not locally isolated) and 7 only by Isolation Forest (locally isolated but not far in Mahalanobis distance). The partial overlap is expected: the two methods capture different geometric notions of anomalousness.

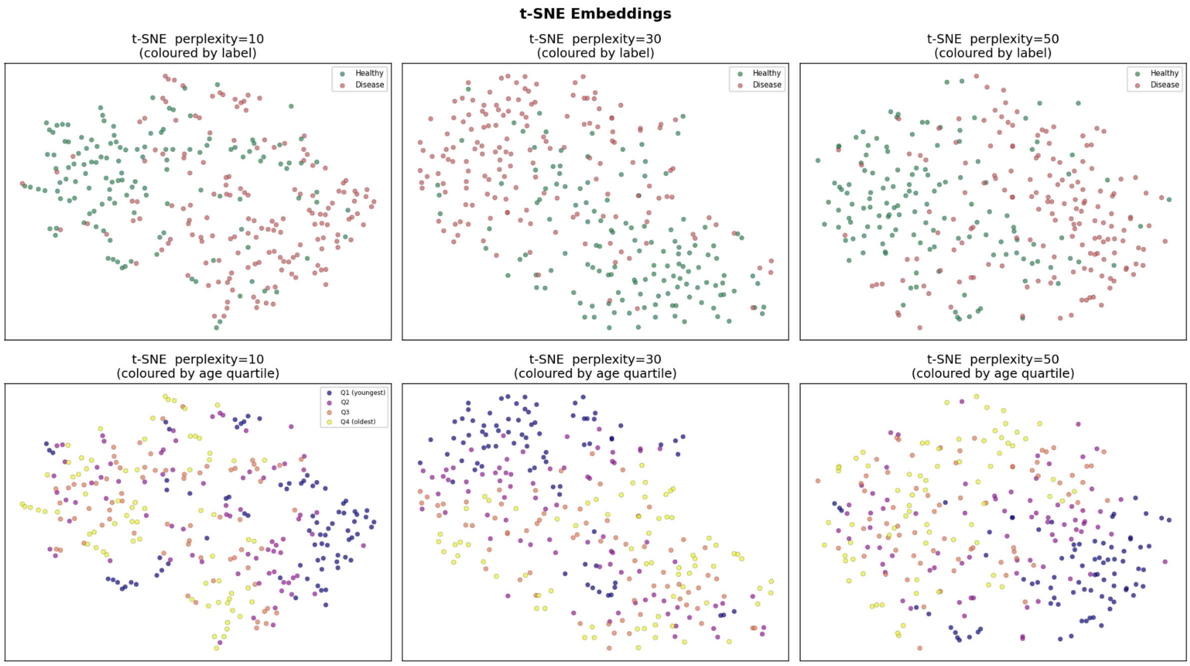
Part 7. Nonlinear Visualisations — t-SNE and UMAP

7.1 Motivation

PCA captures only linear structure. t-SNE and UMAP are manifold learning methods that project data to 2D while preserving local neighbourhood relationships. They can reveal clusters and structure that PCA misses, at the cost of not preserving global distances.

7.2 t-SNE

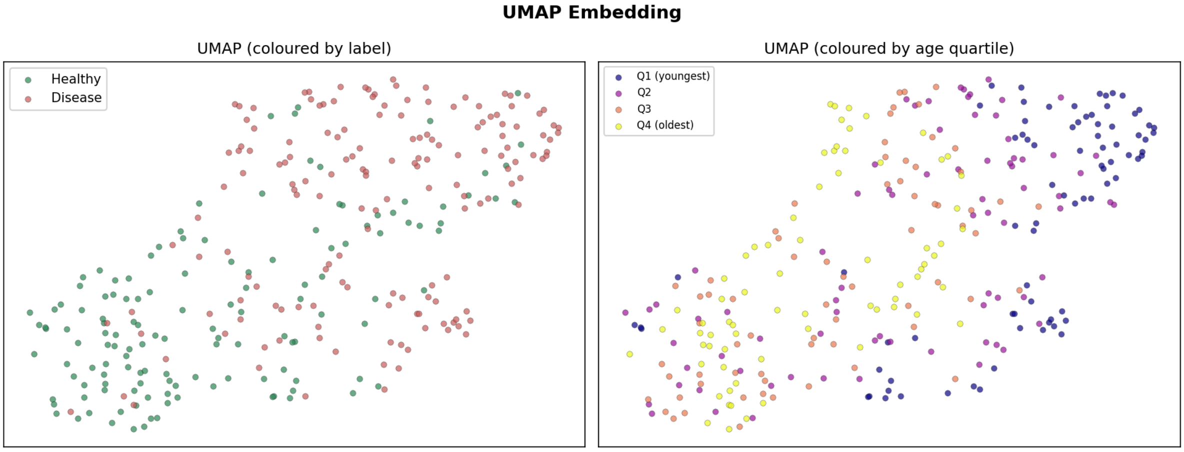
Three perplexity values (10, 30, 50) are used to verify robustness. Perplexity controls the effective number of neighbours considered — low values emphasise local structure, high values more global structure. Consistent patterns across all three confirm the embedding reflects real data geometry rather than algorithm artefacts.



Finding (t-SNE): Across all three perplexity values, the t-SNE embeddings show a consistent pattern: two loosely-separated clusters emerge, broadly corresponding to the healthy (green) and disease (red) classes. The separation is more pronounced at perplexity=30 and 50, where the embedding has smoother structure. At perplexity=10, the embedding is noisier with more fragmented sub-clusters, reflecting the sensitivity to very local structure.

Importantly, the class separation visible here is stronger than in the PCA 2D projection (Section 3.3), confirming that nonlinear manifold structure exists in the data that linear PCA cannot capture. The age-quartile colouring (bottom row) shows no clear age gradient within the clusters, suggesting that the primary separation axis is driven by clinical features other than age.

8.3 UMAP



Finding (UMAP): **UMAP** produces a cleaner, more compact separation than t-SNE. The two classes form two distinct regions with a visible gap between them. UMAP's better preservation of global structure means the inter-cluster distance is more meaningful here than in t-SNE. The age-quartile colouring shows a weak gradient — older patients (Q4, yellow) tend to appear slightly more concentrated in the disease-side region, which is clinically plausible. This trend was not visible in t-SNE, illustrating that UMAP's global structure preservation reveals a relationship that t-SNE's local focus conceals.

Part 8. Fisher Separability & Hubness

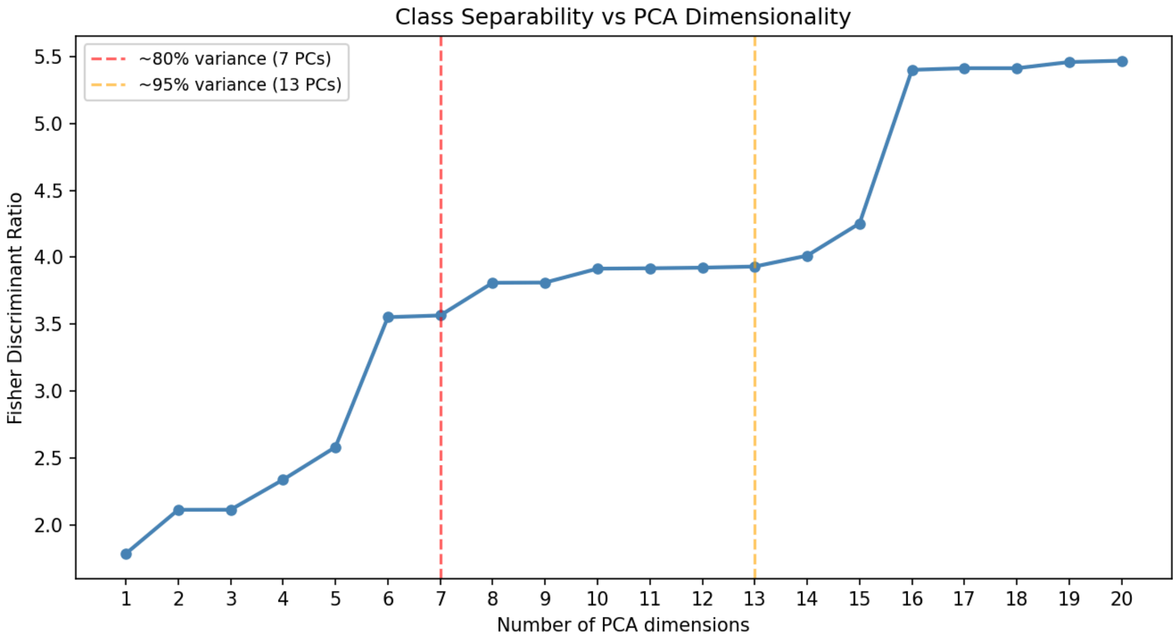
8.1 Motivation

Two complementary analyses quantify class structure as a function of dimensionality.

Fisher Discriminant Ratio (FDR):measures how well the two classes (healthy vs disease) can be separated by a linear boundary, accounting for within-class spread. A higher FDR means the class means are farther apart relative to within-class variance. By computing FDR across PCA subspaces of increasing size, we can see how much each new dimension contributes to separability.

Hubness: in high-dimensional spaces, a small number of points become universal nearest neighbours — they appear in the k-NN lists of many other points despite not being especially close to any particular point. This is a consequence of concentration of measure. The hub count distribution (how many times each point is listed as a neighbour by others) becomes increasingly skewed as dimension grows.

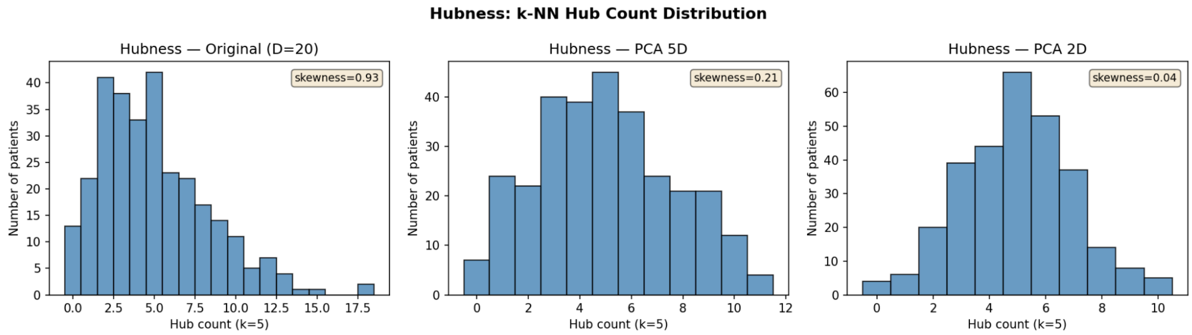
8.2 Fisher Discriminant Ratio vs Dimensionality:



Finding (Fisher Discriminant Ratio): The FDR increases monotonically as more PCA dimensions are added, from 1.78 in 1D to 5.47 in the full 20D space. The curve rises gradually through the first 13 components (reaching 3.93), then jumps more sharply between dimensions 15–16, where PCA components capturing smaller variance contributions still add meaningful class discriminability.

This confirms that class separability is genuinely distributed across many dimensions — no single PCA direction captures it all. The 7-PC subspace (~80% variance) reaches FDR=3.57, while the full 20D space reaches 5.47, indicating that the remaining dimensions contribute meaningfully to linear separability even though they carry little total variance. This aligns with the intrinsic dimensionality findings from Section 4: the data's separable structure lives in roughly 7–9 dimensions.

8.3 Hubness Analysis



Finding (Hubness): The hub count distributions compare the original 20D space against PCA 5D and PCA 2D projections. In the original 20D space, the distribution is right-skewed (skewness=0.93) with a maximum hub count of 18 — some patients appear as k-NN of 18 other patients while 4.4% appear in nobody's k-NN list. This is a signature of the hubness phenomenon in high dimensions.

As dimensionality decreases (5D, 2D), the distributions become more symmetric (skewness drops from 0.93 to 0.04) and the maximum hub count decreases, with fewer zero-hub patients. In low dimensions, the k-NN graph is more evenly distributed. This demonstrates the theoretical prediction: hubness is a high-dimensional phenomenon that diminishes as dimension decreases.

Part 9. Discussion

The analyses collectively reveal a dataset that is moderately high-dimensional but not severely affected by the curse of dimensionality:

Linear structure: PCA requires 13 components for 95% variance, but only 1 eigenvalue exceeds the Marchenko-Pastur noise threshold. The majority of PCA dimensions capture structured noise or correlated features rather than independent signal.

Intrinsic dimensionality: The true geometric dimension is estimated at 7–9 (TwoNN and MLE), well below the ambient dimension of 20 and the PCA threshold of 13. The data lives on a lower-dimensional manifold.

Random projections: The JL lemma holds empirically — projecting to 15 dimensions preserves pairwise distances to within 12% distortion, confirming that most structure is captured in far fewer dimensions than the original 20.

Anomaly detection: Both Mahalanobis distance and Isolation Forest identify approximately 5% anomalous patients. The 8 patients flagged by both methods are the most reliable outlier candidates.

Nonlinear visualisation: t-SNE and UMAP reveal clearer class separation than PCA, confirming that the class boundary is nonlinear and that the manifold structure contains clinically meaningful signal.

Separability and hubness: Fisher separability increases monotonically with dimension, suggesting class information is distributed broadly rather than concentrated. Hubness is present at D=20 (skewness=0.93) but moderate compared to much higher-dimensional datasets.

Part 10 Conclusion

This analysis demonstrates that the UCI Heart Disease dataset occupies an intermediate position in the high-dimensional landscape. Its ambient dimension (20) is modest, but standard high-dimensional phenomena are clearly present: distributed variance, nonlinear manifold structure, a hubness effect, and eigenvalue noise consistent with the Marchenko-Pastur law.

The most parsimonious characterisation of the data is as a 7–9 dimensional manifold embedded in 20-dimensional feature space, with one dominant linear direction (PC1, capturing the cardiac fitness gradient) and approximately 5% anomalous patients. The class boundary between healthy and disease patients is partially linear but benefits significantly from nonlinear embeddings.

Part 11. AI Tools

This project used both Claude and Gemini to think about more analysing methods.
And Claude co-pilot for generating most of the code.